

# Periklis Providis

## EMBEDDED FW & SW ENGINEER

### CONTACT

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Github: [Perry-Pro](#)  
Portfolio: [perry-pro.github.io](#)

### SKILLS

#### PROGRAMMING

[Embedded C] [C/C++] [Python]  
[Assembly]

#### PLATFORMS

[STM32] [ESP32] [RISC-V] [Arduino]  
[PIC] [RPI] [Jetson]

#### PROTOCOLS

[MQTT/TLS] [CAN FD] [SPI] [UART]  
[I2C] [RS485]

#### TOOLS & IDE

[VSCode] [PlatformIO] [GitHub]  
[Matlab] [Simulink] [PSpice] [MPLAB]

#### HARDWARE

[PCB Assembly] [Soldering] [Fusion  
360] [FDM/SLA 3D Printing]

#### OS

[Linux Ubuntu] [MacOS] [Win 10/11]

#### OTHER

[MS Office Tools] [License A2 & B]  
[Google Workspace]

### LANGUAGES

#### English

Professional Working Proficiency

#### Greek

Native / Bilingual

### VOLUNTEER

♥ Platelet Donor

## EMBEDDED FIRMWARE & SOFTWARE ENGINEER

Embedded Firmware & Software Engineer passionate about robotics, microcontroller development, IoT and low-level programming. Hands-on experience across STM32, ESP32, PIC, Arduino and RISC-V MCUs with additional expertise in PCB assembly, HW debugging, 3D modeling & printing. Experienced in taking embedded projects from hardware setup and firmware development through testing and debugging to a working solution.

### WORK EXPERIENCE

#### Embedded Systems Engineer & 3D Designer 07/2023 – 10/2026

Modihive · Thessaloniki, Greece · R&D · [modihive.com](#)

- ESP32 & RISC-V firmware in C/C++; protocols: MQTT/TLS, CAN2, CAN FD, SPI, UART, I2C, RS485
- 5G IoT & Raspberry Pi 4 project development; NVIDIA Jetson flashing and setup
- PCB assembly, component soldering, HW/FW/SW debugging; Python utility scripts and automation
- GitHub version control, VSCode + PlatformIO; hardware stock tracking (spreadsheets & docs)
- 3D design (Fusion 360) & printing (Bambu Lab); environments: Windows, Linux Ubuntu, macOS

#### Medical/Dental Machine Specialist 11/2022 – 06/2024

Apostolides Healthcare Technologies · Thessaloniki, Greece, [apostolides.gr](#)

- 3D printing for dental applications — [Formlabs](#)
- CBCT radiology machines — [Cefla](#) & [Planmeca](#)

#### Refrigeration Technician 06/2016 – 04/2019

Real Energy · Kalamaria, Thessaloniki, Greece, [realenergy.gr](#)

### CERTIFICATES – EMBEDDED

#### Advanced C Programming Course **[IN PROGRESS]**

Master C — apply for Embedded / Real-Time C roles

#### Mastering MCU: Timers, PWM, CAN, Low Power (MCU2) **[IN PROGRESS]**

STM32 Timers, CAN, RTC, PWM & Low Power via HAL APIs

#### Mastering MCU & Embedded Driver Development 04/2024 – 07/2024

28.5h — Bare metal: STM32 GPIO, I2C, SPI, USART from scratch [link](#)

#### Microcontroller Embedded C: Absolute Beginners 02/2024 – 03/2024

16.5h — Foundation Embedded C with STM32 [link](#)

#### Arduino e-Learning Academy — CollegeLink 10/2022

25h — 4 Smart Home projects on Tinkercad Autodesk [link](#)

### EDUCATION

#### MSc — in Mechatronics 2026 – Present

University of Western Macedonia, Dept. of Electrical and Computer Engineering, [mechatronics.uowm.gr](#)

Embedded Systems, Microprocessors, Industrial Robotics, Control Systems, IoT

#### BSc — Department of Automation 2016 – 2022

International Hellenic University, Thessaloniki, [autom.teithe.gr](#)

- Final grade: 6.71 "Very Good" [link](#)
- Thesis (10/10): "3D Printed Robotic System" [Drive link](#) [Github link](#)
- **Mechanical:** 5-DOF robotic arm + gripper, fully custom 3D modeled and FDM printed
- **Electronics:** Arduino Uno, custom electrical circuits, HC-06 Bluetooth module
- **Software:** C++ FW for arm control, companion Android app for real-time wireless operation
- **Impact:** Retained by the department lab as a teaching artifact, with the aim of forming the basis of a dedicated course on robotic systems

#### PIC Microcontroller Course 2022 – 2023

Lamprou Konstantinos, MSc · Remote · [picmicrocontroller.gr](#)

- Embedded C Language, PIC18xx / PIC24xx architecture

### PERSONAL PROJECTS

#### 3D Printed Bluetooth Speaker 07/2022

- Self-built from scratch: research, 3D modeling, 3D printing, and circuit design [link](#)

### REFERENCE LETTERS

#### Ph.D Tsagaris Apostolos 10/2022

Thesis supervisor — Mechatronics, PLC & SCADA, CIM [link](#)

#### Ph.D Manolakis Dimitrios 10/2022

Programming in C/C++ [link](#)

#### Ph.D Nikolaidis Nikolaos 10/2022

Microcomputers & Microcontrollers [link](#)